EXECUTIVE DECISION

Bring AI capacity into engineering

A pilot of **six weeks** to reduce variable spending, retain sensitive context and maintain alternatives.

Local first.
Cloud when it adds value.

Scope: code, backend, UI and tests. Hybrid means local + cloud.

Planning proposal · 16 October 2026

01 / THE PROBLEM

Per-token consumption changes the bill

The team report points out users with spending of **$2,000–10,000**. We assume monthly amounts per professional here; verify against invoices.

US$ 10 thousand

5 people × US$ 2 thousand/month

US$ 50 thousand

5 people × US$ 10,000/month

This is not a market average. Cache, repeated context and parallel agents must appear in the baseline.

User-provided account · Arithmetic scenarios; no audited invoice

02 / OPPORTUNITY

Compact models deserve qualification

Benchmark	Qwen3.6 · 27B	Qwen3.8 · 27B
Terminal Bench 2.1	63.4	**73.0**
SWE-bench Pro	53.5	**61.7**
NL2Repo	36.2	**42.3**

The same nominal parameter count, with a higher published result. Validate quantization and internal tasks; the dataset was not isolated as the sole cause.

Qwen · evaluation of the authors themselves

03 / BUSINESS EVIDENCE

Hybrid savings depend on the task

Task	Cloud → hybrid	Change
Ktor mapper	$0.339 → $0.134	−60.5%
TypeScript rename	$0.227 → $0.121	−46.7%
Spring field	$0.092 → $0.164	+78.3%

Local Qwen on Mac + remote Sonnet. Select task families using accepted-task cost; these are not monthly team savings.

04 / PERFORMANCE

Start with 32, 48 and 64 GB too

Mac	Model / runtime	Decode tok/s
M1 Pro · 32 GB	35B MoE Q4 · llama.cpp	25.4
M4 Pro · 48 GB	27B Q4 · Rapid-MLX	15.2
M4 Pro · 64 GB	27B Q4 / Q8 · oMLX + MTP	20.8 / 16.6

Reuse workstations before purchasing. Different protocols; the 32 GB test records host swap. Memory alone does not determine speed or quality.

M1 Pro 32 GB · M4 Pro 48 GB · M4 Pro 64 GB

05 / ARCHITECTURE

Preserve context. Escalate according to policy.

Repository + IDE

Identity and scope per project

→

Local model + isolated tools

Search, editing, testing and controlled logs

→

Automated validation + review

Build, tests, contracts and UI

Authorised exceptions: the gateway minimizes context and calls the cloud. Data prohibited from leaving stays local.

Proposed architecture · fallback never replaces authorization to send data

06 / SECURITY

Internet access with enforced boundaries

Isolated agent worker → authenticated local model → disposable tests.

Documentation through a restricted broker. Cloud export through a separate authorization gate.

No production credentials, broad host mounts or unrestricted shell egress. Review generated workflows and hooks as well as code.

Docker security model · Sandbox runtime

Qualify OMP, OpenCode and Aider

Layer	Decision
MLX / llama.cpp	Private, authenticated inference gateway
Agent	Restricted effective tool policy
Execution	OS/VM boundary and controlled network
Acceptance	Independent tests and human review

Open-weight deployment is manageable. The agent name alone is not a security guarantee.

OpenCode · OMP · Aider

07 / SECURITY VALIDATION

Prove containment before expansion

Test malicious repository instructions, web redirects, synthetic-secret export, dependency scripts and access outside the worktree.

Verify that project configuration cannot weaken corporate policy and that cloud fallback cannot bypass data restrictions.

Evidence: denied operations, network traces, accepted patches and bounded execution. Physical side channels remain a specialist appendix.

Isolation layers and documented limitations

Use the cloud to improve reusable local workflows

Cloud optimizer proposes a task-specific harness. Independent tests select a version. The local model executes recurring work.

Version instructions, retrieval, tools and bounded recovery; preserve rollback and untouched acceptance tests.

SkillOpt supports optimizing skills without changing weights. Coding savings must still be measured on the actual repository.

Microsoft SkillOpt

09 / WHAT CHANGES AT WORK

Start with verifiable tasks

Workstream	Initial application	Gate
Backend	APIs, SQL and Delimited Refactoring	Contracts + tests
QA	Cases, fixtures and regressions	Execution + useful coverage
UI	Components and visual corrections	Captures + accessibility + E2E
Legacy	Explanation and changes per module	Behavior characterization

Accepting code requires evidence of execution, even when the agent says it has finished.

Proposed engineering plan; internal rates will be measured during the pilot

10 / INVESTMENT

US$ 41,400 to qualify the approach

Component	Assumption	US$
Workstations	5 × 5,000	25,000
Integration	80 h × 100	8,000
Security + validation	30 h × 100	3,000
Contingency	15%	5,400

Estimated additional operation: **US$ 500/month**. Configurations and prices still require quotations.

Planning scenario, not a commercial quotation from a supplier

The avoided portion determines payback

API/person/month	Team savings / month	Payback
US$ 2,000	US$ 4,500	9.20 months
US$ 5,000	US$ 12,000	3.45 months
US$ 10,000	US$ 24,500	1.69 month

5 users · 50% of invoice avoided · $500/month of operation · full investment of US$ 41,400.

Sensitivity: in the first scenario, avoiding only 20% leads to a return to 27.6 months.

Own calculation · additional rework must be discounted · calculator in the proposal

12 / PILOT

Six weeks, with criteria defined in advance

1

Baseline

Invoices, tasks, quality and review

2

Implementation

Configuration, isolation and controls

3–5

Comparison

Real tasks, current flow versus hybrid

6

Decision

Savings, quality, security and capacity

Proposed targets: ≥30% total savings; ≤5 pp loss of acceptance; ≤10% increase in cycle; no exposure in defined tests.

Criteria for negotiation; these are not results already achieved

13 / RECOMMENDATION

Buy evidence before scaling

Approve the pilot, confirm the baseline and make the expansion result dependent.

Cost per accepted task.
Data governed by policy.
Versions under control.

Platform + Security + FinOps + Engineering/QA share the decision.

Next proposed step · no purchase, hiring or automatic publication

From decision to detail

Full research

Business cases, Macs tables 16–512 GB, MLX/llama.cpp, SSD, experts and security.

Open Report →

Investment proposal

Problem/Context, Solution, Required Investment and sensitivity calculator.

Open Proposal →

Sources appear in the footnotes and documents. Arrow keys navigate between slides; “Print / PDF” exports the entire presentation.

Offline package · 16 October 2026

TECHNICAL SUPPORT / SSD

Larger models on the same workstations

Mac / model	SSD path	Decode tok/s
32 GB M1 Max · Flash-Next	MoEspresso · mainly IQ2_K	12–15
48 GB M5 Pro · Flash-Next	Slotstream · 4-bit	~12
64 GB M5 Pro · Flash-Next	llama.cpp fork / Slipstream	27.6 / 35.0–37.5
64 GB M5 Pro · DeepSeek V4.1	WARP · CPU	3.77 / 3.71

Authors’ results, without a common protocol. MoEspresso changes routing. WARP measures 64 / 200 tokens; it does not establish the 20–30 target.

MoEspresso · Slotstream · 64 GB · WARP · Slipstream

DECISION SUPPORT / INVESTMENT

Smaller purchase, the same economic discipline

Compare reuse with **US$ 2500–5000** workstations: five units total US$ 12500–25000 in hardware. Planning amounts; obtain configuration and SSD quotations.

The home account of ~60 tokens/s uses a 32 GB VRAM GPU + 64 GB RAM + SSD; logs and full configuration are still unavailable. It is not a Mac benchmark.

Measure four paths: cloud, generic local, local with harness, and local with harness + cloud. Approve based on net savings, quality, time and data containment.

Pilot proposal · account without independent reproduction · TCO includes integration and operation